

Multiple Choice (10 marks)

1. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
2. A special tool is necessary for entering the network which is _____ that helps the anonymous internet users to access into the Tor's network and use various Tor services.
 - (a) Opera browser
 - (b) Firefox
 - (c) Chrome
 - (d) Tor browser
3. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
4. What is penetration testing?
 - (a) A procedure for testing libraries or other program components for vulnerabilities
 - (b) Whole-system testing for security flaws and bugs
 - (c) A security-minded form of unit testing that applies early in the development process
 - (d) All of the above
5. 1. _____ framework made cracking of vulnerabilities easy like point and click.
 - (a) .Net
 - (b) Metasploit
 - (c) Zeus
 - (d) Ettercap

True / False (10 marks)

Answer True or False

6. True or False: A Base Transceiver Station (BTS) is similar to an Access Point in 802.11, used by mobile operators for signal coverage.
7. True or False: A buffer overflow occurs when a pointer is used to access memory not allocated to it.
8. True or False: The one-time pad is considered a perfectly secure encryption scheme when used correctly, as it provides information-theoretic security.
9. True or False: A MAC with a fixed tag length of 5 bits is vulnerable to brute-force attacks, allowing an attacker to guess valid tags.
10. True or False: The sub key length at each round of DES is 56 bits.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to get an item of a tuple.
12. Write a function to find the maximum profit earned from a maximum of k stock transactions

End of Paper